Linear Algebra Lecture Notes

Chapter 1

Basic definitions:

- Linear equation
- System of linear equations
- Solution
- Solution set
- Equivalent linear systems
- Matrix
- Coefficient matrix
- Augmented matrix
- Size of matrix

Finding the solution set of a system of two linear equations in two variables is easy because it amounts to finding the intersection of two lines.

A typical problem is :

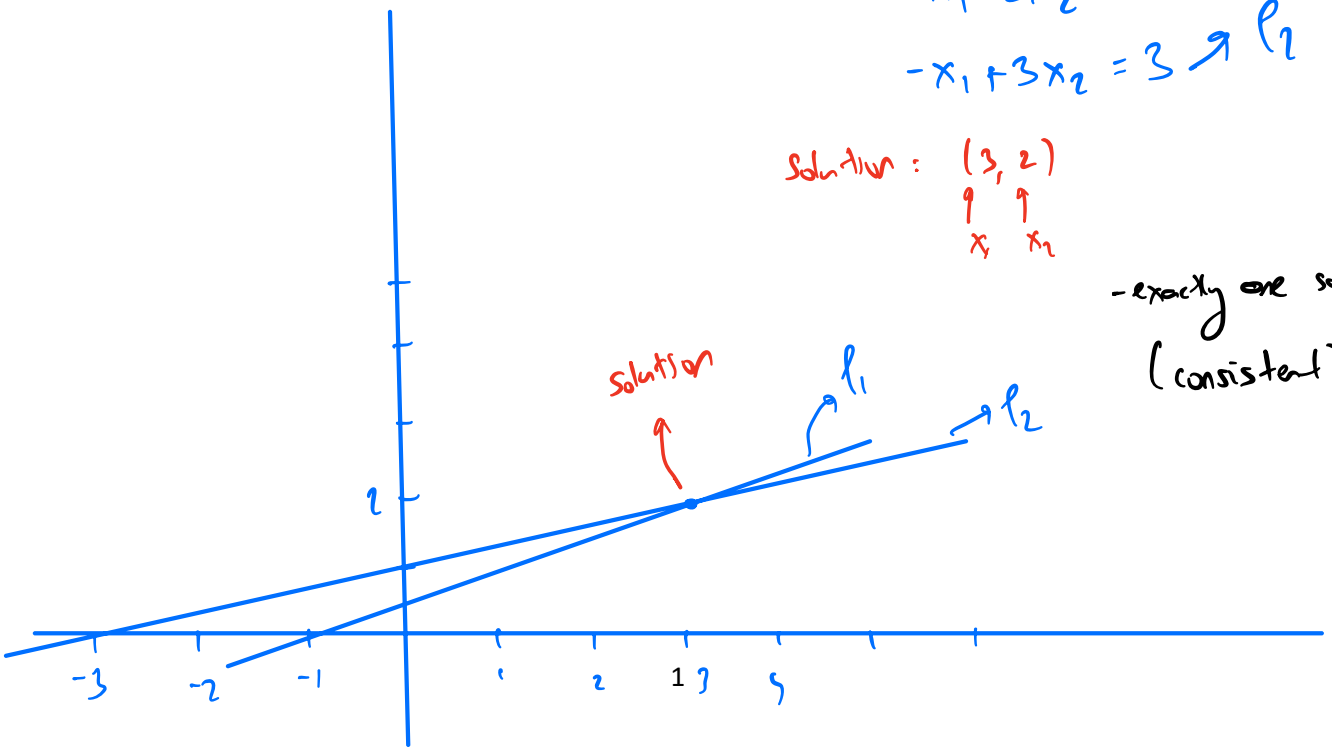
Example:

$$x_1 - 2x_2 = -1 \quad (l_1)$$

$$-x_1 + 3x_2 = 3 \quad (l_2)$$

Solution : $(3, 2)$ (x_1, x_2)

- exactly one solution (consistent)

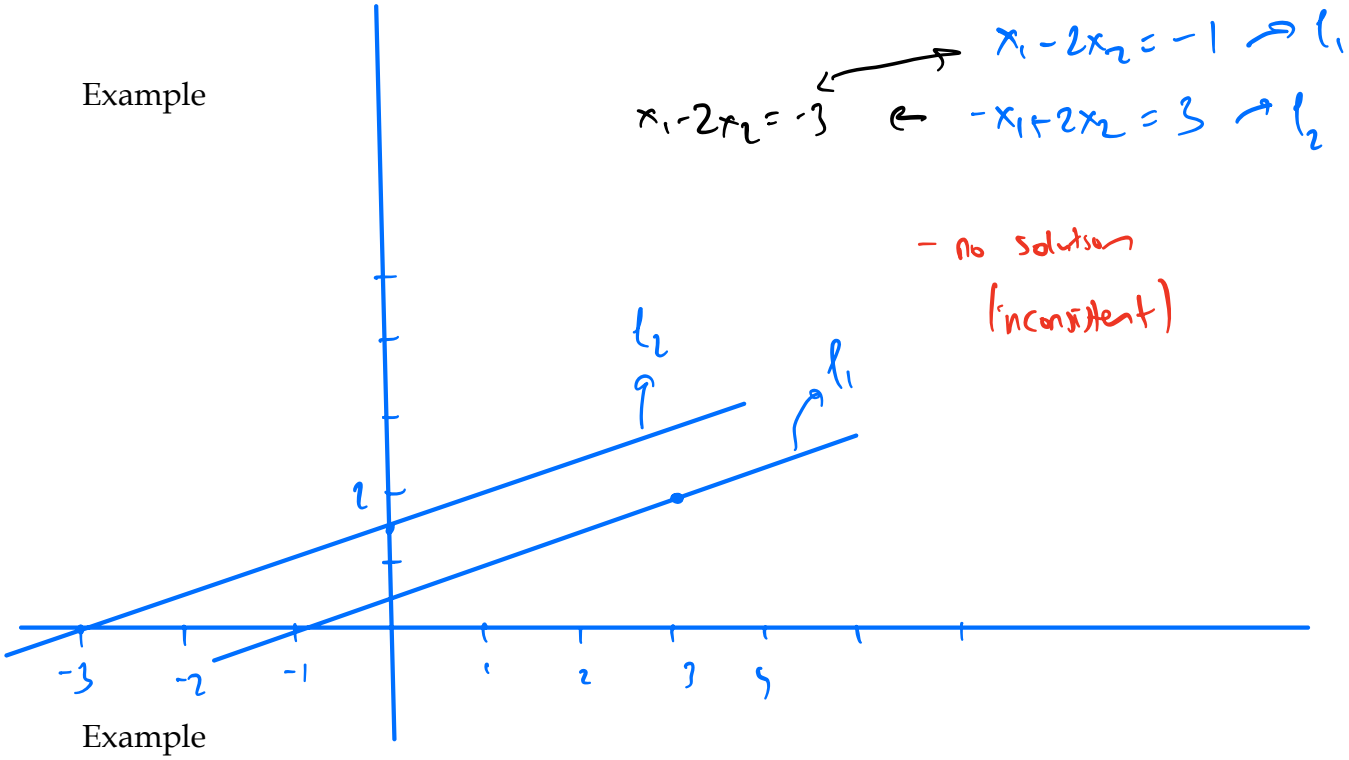


A pair of numbers (x_1, x_2) satisfies both equations in the system if and only if the point (x_1, x_2) lies on both l_1 and l_2.

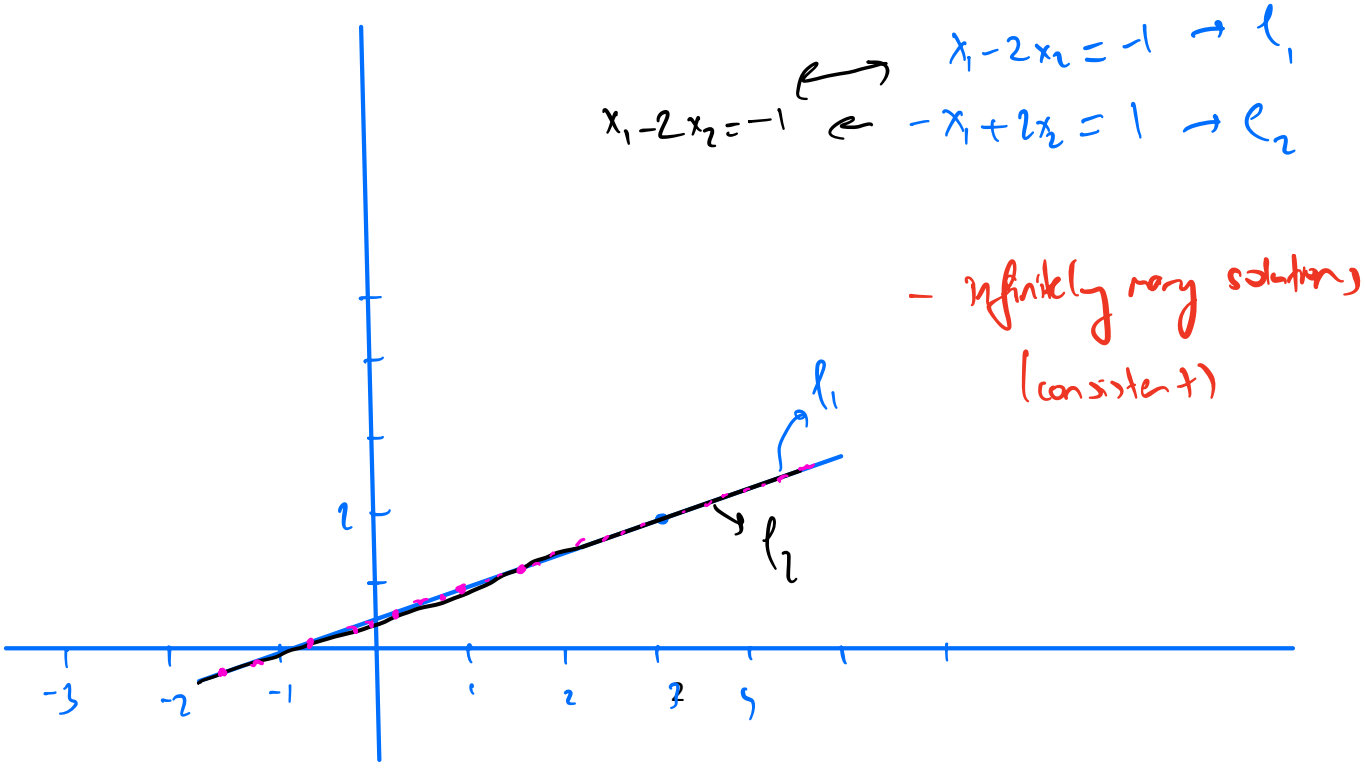
Two lines need not intersect in a single point

- They could be parallel, or
- They could coincide and hence intersect at every point on the line

Example



Example



Three basic operations are used to simplify a linear system.

1. Replace one equation by the sum of itself and a multiple of another equation,
2. Interchange two equations,
3. Multiply all the terms in an equation by a nonzero constant

Rules 2 and 3 are easy to justify. Let's give examples towards a better understanding of Rule 1.

E_1 $x_1 - 2x_2 = -1 \to l_1$

E_2 $-x_1 + 3x_2 = 3 \to l_2$

E_3 $-x_1 + 4x_2 = 5 \to l_3$

E_4 $2x_1 + 3x_2 = 12 \to l_4$

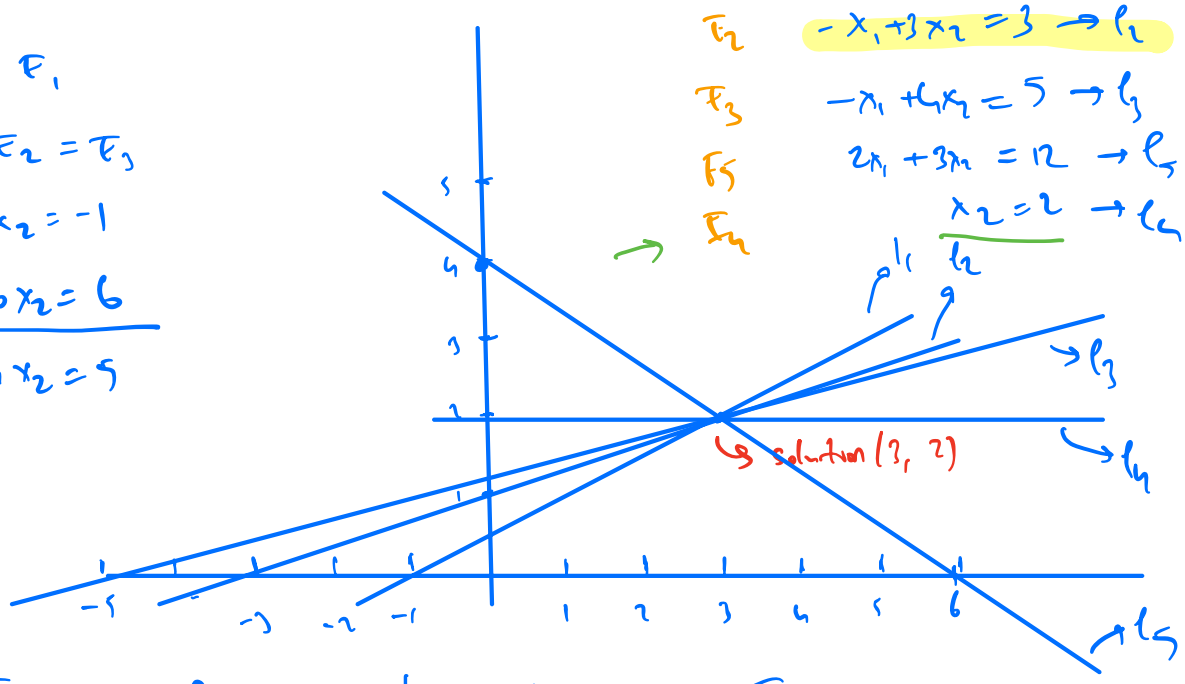
E_5 $x_2 = 2 \to l_5$

Replace E_1 by $E_1 + 2E_2 = E_3$

E_1: $x_1 - 2x_2 = -1$

$2E_2$: $-2x_1 + 6x_2 = 6$

E_3: $-x_1 + 4x_2 = 5$



obtain $E_5 = 3E_1 + 7E_2$

$3E_1$: $3x_1 - 6x_2 = -3$

$7E_2$: $-7x_1 + 21x_2 = 21$

E_4: $2x_1 + 3x_2 = 12$

obtain $E_5 = E_1 + E_2$

E_1: $x_1 - 2x_2 = -1$

E_2: $-x_1 + 3x_2 = 3$

E_5: $x_2 = 2$

So, adding one row a multiple of another row does not change the solution set. If it is wisely done we can eliminate one variable from an equation.

(E_1, E_2) (E_2, E_3) (E_3, E_4) (E_4, E_5)
(E_1, E_3) (E_2, E_4) (E_3, E_5)
(E_1, E_4) (E_2, E_5)
(E_1, E_5)

are all equivalent linear systems.

You may use any two to solve the system.

Solving a linear system

- The basic strategy is to replace one system with an equivalent system (i.e. one with the same solution set) that is easier to solve.
- Use the x_1 term in the first equation of a system to eliminate the x_1 terms in the other equations. Then use x_2 term in the second equation to eliminate the x_2 terms in the other equations, and so on, until you finally obtain a very simple equivalent system of equations.

Example:

$$\begin{aligned} E_1 &\to x_1 - 2x_2 + x_3 = 0 \\ E_2 &\to 2x_2 - 8x_3 = 8 \\ E_3 &\to 4x_1 + 5x_2 + 9x_3 = -9 \end{aligned}$$

$$\begin{matrix} & x_1 & x_2 & x_3 & b \\ E_1 \to & 1 & -2 & 1 & 0 \\ E_2 \to & 0 & 2 & -8 & 8 \\ E_3 \to & 4 & 5 & 9 & -9 \end{matrix}$$

We want to keep x_1 in the first equation and eliminate it from the other equations.

$$\begin{aligned} 4E_1: \quad & 4x_1 - 8x_2 + 4x_3 = 0 \\ E_3: \quad & -4x_1 + 5x_2 + 9x_3 = -9 \\ \hline 4E_1 + E_3: \quad & 0 - 3x_2 + 13x_3 = -9 \end{aligned}$$

→ write this in the third equation.

$$\begin{aligned} E_1 \to \quad & x_1 - 2x_2 + x_3 = 0 \\ E_2 \to \quad & 2x_2 - 8x_3 = 8 \\ E_3 \to \quad & -3x_2 + 13x_3 = -9 \end{aligned} \longrightarrow \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{bmatrix}$$

Multiply E_2 by $1/2$ in order to obtain 1 as the coefficient for x_2.

$$\begin{aligned} E_1 \to \quad & x_1 - 2x_2 + x_3 = 0 \\ E_2 \to \quad & x_2 - 4x_3 = 4 \\ E_3 \to \quad & -3x_2 + 13x_3 = -9 \end{aligned} \longrightarrow \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{bmatrix}$$

Use the x_2 in E_2 to eliminate $-3x_2$ in E_3.

Therefore, write $3E_2 + E_3$ in E_3.

$3E_2$: $3x_2 - 12x_3 = 12$

E_3: $-3x_2 + 13x_3 = -9$

$3E_2 + E_3$: $x_3 = 3$

E_1 $x_1 - 2x_2 + x_3 = 0$

E_2 $x_2 - 4x_3 = 4$

E_3 $x_3 = 3$

$$\rightarrow \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Eventually we want to eliminate the $-2x_2$ term from E_1.

But it is more efficient to use x_3 in E_3 first to eliminate the $-4x_3$ and x_3 term in E_2 and E_1, respectively. That is,

$4E_3 + E_2$: $4x_3 = 12$

$+ \quad x_2 - 4x_3 = 4$

$x_2 = 16$

$-E_3 + E_1$: $-x_3 = -3$

$+ \quad x_1 - 2x_2 + x_3 = 0$

$x_1 - 2x_2 = -3$

So, the linear system becomes:

E_1 $x_1 - 2x_2 = -3$

E_2 $x_2 = 16$

E_3 $x_3 = 3$

$$\rightarrow \begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Eliminate $-2x_2$ from E_1.

$$
\begin{array}{lrcl}
2E_2 + E_1: & x_1 - 2x_2 & = & -3 \\
+ & 2x_2 & = & 32 \\
\hline
& x_1 & = & 29
\end{array}
$$

So, the equivalent linear system is:

$$
\begin{aligned}
x_1 &= 29 \\
x_2 &= 16 \\
x_3 &= 3
\end{aligned}
\quad \rightarrow \quad
\begin{bmatrix}
1 & 0 & 0 & 29 \\
0 & 1 & 0 & 16 \\
0 & 0 & 1 & 3
\end{bmatrix}
$$

Two fundamental questions about a linear system:

1. Is the system consistent, that is, does at least one solution exist?
2. If a solution exists, is it the only one, that is, is the solution unique?

Example:

$$x_2 - 4x_3 = 8$$
$$2x_1 - 3x_2 + 2x_3 = 1$$
$$5x_1 - 8x_2 + 7x_3 = 1$$

The augmented matrix is

$$\begin{bmatrix} 0 & 1 & -4 & 8 \\ 2 & -3 & 2 & 1 \\ 5 & -8 & 7 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 5 & -8 & 7 & 1 \end{bmatrix} \rightarrow \quad F_{1/2} \rightarrow \begin{bmatrix} 1 & -3/2 & 1 & 1/2 \\ 0 & 1 & -4 & 8 \\ 5 & -8 & 7 & 1 \end{bmatrix}$$

$$\rightarrow \quad -5F_1 + F_3 \rightarrow \begin{bmatrix} 1 & -3/2 & 1 & 1/2 \\ 0 & 1 & -4 & 8 \\ 0 & -1/2 & 2 & -3/2 \end{bmatrix} \rightarrow \quad F_2/2 + F_3 \rightarrow \begin{bmatrix} 1 & -3/2 & 1 & 1/2 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

The last row means that

$0x_1 + 0x_2 + 0x_3 = 5/2$ i.e. $0 = 5/2$.

The system has a built in contradiction because no x values satisfy this equation. Since our original system and last system are equivalent, we can say that the original system is inconsistent (no solution).